SEVA INNOVATION CHALLENGE 2026 • MAHARASHTRA ZONE

NyaySaakshya

Know what an email-traffic capture proves, and what it cannot.

A working prototype for forensic triage of saved email-traffic captures. Each issue it reports cites the packets behind it. When a capture cannot support a score, NyaySaakshya says so. It does not detect fraud or identify suspects.

Applicant	**Siddhant Hiralal Patil**
Application	Individual (team size 1)
Sector	Law and Order
Sub-theme	Cyber-fraud, digital extortion and financial-crime intelligence
Stage	**Working Prototype**: lab-validated, no external pilot

Real output of the current build, synthetic lab captures



Enough evidence: a score, with the rule, standard and packet it rests on



Not enough evidence: no score, and the reason is stated

Cropped from the working console; not edited.

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The Problem

A capture can show less than it seems to. Reading too much into it can undermine the finding built on it.

Three ways a capture can mislead an investigator

What the capture shows	Easy to conclude	What it actually supports
It ends before the TLS handshake finishes	No encryption was used	Incomplete: the capture stops before the answer is known
The server never offers STARTTLS	Someone stripped the encryption	Ambiguous: genuine non-support and stripping look identical in the bytes
TLS 1.3 encrypts the server certificate	The certificate is invalid	Not observable in a passive capture

Why it matters

A conclusion the evidence cannot support is a weak foundation for a case.

Who is affected

Cyber-cell and forensic investigators and their supervisors. Proposed first users: a Maharashtra cyber cell or forensic lab; no partner has agreed.

The gap today

Packet tools show packets. Deciding what is safe to conclude is manual and hard to repeat.

Not yet known

How often investigators meet such captures. None consulted yet.

Context only: PIB (7 Oct 2025) cites over 17.82 lakh complaints in the national CFCFRMS cyber-fraud system; period not stated. It is not email-specific, and we make no claim about how many cases involve email.

3 Our Solution & Innovation

Every fact is labelled by how strongly the capture supports it. If it cannot support a score, none is given.

Capture — Web upload of a PCAP or PCAPNG; SHA-256 recorded

TShark parse — Read-only. No live capture, no decryption

Sessions — Sessions rebuilt; every fact gets an evidence state

Rules — 19 versioned rules cited to RFC or NIST. No AI in verdicts

Posture — A score, or Not scored when evidence is thin

Reports — JSON, HTML, PDF with packet citations

Six evidence states

OBSERVED
Seen directly in the captured bytes

INFERRED
Deduced; the basis is stated

UNKNOWN
Too little evidence to decide

AMBIGUOUS
More than one reading fits

INCOMPLETE
Capture is cut off at that point

NOT OBSERVABLE
Cannot be seen passively

Navy: can support a conclusion. **Orange**: cannot.

One finding, traced to its packets

Source	Capture SHA-256 97b2b61a...; stream #0, 10.0.0.6 to 10.0.0.80:587
Evidence	Frame #7: authentication activity = True (OBSERVED); TLS transition = False (OBSERVED)
Rule	SEC-PLAIN-001, citing NIST SP 800-52 Rev. 2 section 3.1
Finding	Authentication activity without TLS protection: CONFIRMED

From a real run on a synthetic lab capture.

Withholding a score is deliberate. If no session can be assessed, or fewer than half yield a conclusion, the result is Not scored, with the reason. Blindness is never reported as safety.

How it differs: Wireshark, Zeek and NetworkMiner expose packets and fields. The documentation we reviewed does not describe per-fact evidence states (not proof they lack them). A design strength, not uniqueness.

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Feasibility & Implementation

A working prototype, tested internally, and an honest list of what a pilot would still need.

Current stage: Working Prototype (lab-validated; no external pilot)

Idea — Proof of Concept — ✓ Working Prototype — Field Pilot — Ready to Deploy

Built and tested (internal)

- Python and TShark analysis, React console, FastAPI service, SQLite
- 1,521 automated tests
- 98 controlled lab captures: 10 from Postfix and Dovecot software in a lab, 6 OpenSSL handshakes, 82 crafted in-project
- Repeat runs gave the same result (98 of 98 captures)
- Failure found and fixed: on 44 cut-off streams the engine said "no TLS" 41 times; now once (an arguable case)

Lab validation only: no real-world traffic and no accuracy benchmark.

Not built yet

- Login and roles
- Audit log
- Encrypted storage, backups, retention
- Fix for a database-concurrency defect
- Enforced analysis time limit
- Sandboxed parsing of untrusted files

Not production-ready. The demo is local, with synthetic captures only.

Proposed path (not committed)

- **0 to 3 months:** close the gaps; write a threat model
- **3 to 6 months:** technical pilot on synthetic captures
- **6 to 12 months:** supervised single-unit pilot, only if a department agrees; real case data only once authorised and prerequisites are met

Depends on security mentorship, a willing unit and its legal approval for real captures. Cost not priced; a technical pilot needs no cloud service or GPU.

5 Impact & Scalability

Fewer unsupported conclusions in cyber-fraud cases: the expected benefit, to be measured in a supervised pilot.

Public value (expected, not yet measured)

- Investigators and supervisors can see what a capture supports and which packets show it
- Findings can be re-checked: repeat runs gave the same result
- Fewer conclusions that the evidence cannot bear
- Fits SDG 16, target 16.a: strengthening institutions that combat crime

Adoption path (proposed)

1. Hardened local build
2. One supervised unit: a cyber cell or forensic lab
3. Measured results, then decide on more units

Likely owner: Home Department (police cyber cells); no office approached yet. Each unit would run it locally: a design intention, not built or validated.

How success would be measured

- Time to a first assessment
- Time to a finished report
- Findings with usable packet citations
- Handling of unresolved evidence
- Repeat-run agreement
- Supervisor confidence

Before: current practice, measured in the pilot. After: with NyaySaakshya. No targets claimed. The existing cost it targets is analyst time, to be measured.

What the Challenge would unlock

Security mentorship and one supervised pilot would turn a lab-validated prototype into a measured public-sector trial of whether it helps investigators avoid unsupported conclusions.

Not claimed: beneficiaries reached, time or cost saved, jobs. None has been measured.

Sustainability: no revenue model is decided. It builds on open-source components (TShark, Python, FastAPI, React); NyaySaakshya's own licence is not yet chosen.

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Team/Individual & Support Required

One applicant with a working prototype, and the specific help needed to reach a supervised pilot.

SP

Siddhant Hiralal Patil

Applicant and team lead

Individual application (team size 1)

No other team member, mentor, partner or funder is named.

Next milestone (proposed, about 3 months)

A hardened local pilot-candidate build: login and roles, audit log, encrypted storage, backups and a written threat model.

Why this help

The core analysis is built. What blocks a pilot is security engineering and access to a supervised unit.

Demonstrable now

A local demo on synthetic captures: a supported finding, a withheld score and a report export.

Support we are seeking

- [x] **Security mentorship**
  Review the threat model and hardening plan for sensitive evidence
- [x] **A supervised pilot unit**
  Cyber cell or forensic lab; synthetic data first. No partner yet
- [x] **Testing, certification and IP guidance**
  Choose a licence; understand evidence-handling expectations
- [x] **Funding for a secure pilot build**
  Login, audit log, encrypted storage and backups
- [ ] Incubation and workspace
  Not requested

Thank you | Siddhant Hiralal Patil